

Summary and Recommendations

Objective:

This analysis investigates factors influencing customer churn, focusing on contract types, payment methods, tenure, and demographic attributes. The goal is to identify key drivers and recommend strategies to improve retention.

Key Findings:

1. **Churn by Contract Type:**
 - Month-to-month contracts show a churn rate of **42%**, significantly higher than **11%** for yearly contracts and **3%** for two-year contracts.
 - **Implication:** Long-term contracts strongly reduce churn and enhance stability.
2. **Churn by Payment Method:**
 - Electronic check users have the highest churn rate at **45%**, while credit card, bank transfer, and mailed check users average **15-18%**.
 - **Implication:** Addressing convenience and trust concerns can encourage more stable payment methods.
3. **Churn by Tenure:**
 - Customers with less than a year's tenure exhibit a **50% churn rate**, which decreases to **15%** for tenures over three years.
 - **Implication:** Initial engagement within the first year is critical for retention.
4. **Churn by Internet Service Type:**
 - Fiber optic users have a **30% churn rate**, higher than the **20%** for DSL users.
 - **Implication:** Investigating fiber optic dissatisfaction may reveal service quality issues.
5. **Senior Citizen Churn:**
 - Senior citizens have a **41% churn rate**, compared to **26%** for non-senior customers.
 - **Implication:** Tailored retention strategies for seniors can reduce churn in this demographic.

Visualizations & Insights:

- Bar charts and line graphs highlight disparities in churn by contract type, payment method, and tenure.
- Visual trends reveal a sharp decline in churn with longer tenures and show electronic check users are almost three times more likely to churn.

Recommendations:

1. **Promote Long-Term Contracts:** Provide incentives to encourage customers to commit to longer contracts.
2. **Address Payment Method Concerns:** Encourage customers to shift from electronic checks to credit cards or bank transfers through campaigns highlighting convenience and security.
3. **Focus on Early Engagement:** Develop loyalty programs targeting customers in their first year to reduce churn.
4. **Improve Service Quality:** Investigate customer dissatisfaction with fiber optic services to enhance retention.
5. **Senior Citizen Programs:** Create personalized offers or services for senior customers to increase satisfaction and loyalty.